

CHAPTER 5

Blockchain for intrusion detection systems

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1. Introduction

Nowadays, much work is done online, giving rise to cyber-attack. These attacks have become advanced and very complicated. To overcome these problems and detect suspicious or malicious activities in a short time, intrusion detection systems (IDSs) are being used and implemented in different fields (such as health, banking, and education). If such an activity occurs, it is collected by security information and event management systems. In some cases, these IDS are capable of responding to such activity. So, to understand the importance of blockchain in an IDS, we first need to understand what intrusion detection actually is and why we need it. An important step is to select a suitable IDS technology, which is basically of two types: network intrusion detection (NIDS) or host-based intrusion detection (HIDS). The IDS analyzes networks for any malicious activity. It is positioned on network telecommunications like Ethernet and monitors the contents of packets passively. It looks for volume, source, and destination of the traffic, and it verifies the packets that should be present in the network and the ones that should not be present and then alerts the user upon suspicious activity (Kumar and Singh, 2020). It monitors important operating system files. It looks for resources each program accesses, authentication tables, changes that are made in memory at particular locations, RAM storage state information, and log-related files. It also monitors dynamic behavior and state of the system and compares that with what it actually should be.

Detection approaches are divided into two parts: signature-based IDS and anomaly-based IDS.

1.1 Signature-based IDS

This is used to detect threats known to us. It relies on a preprogrammed list of known threats. This technique looks at specific patterns like malicious byte sequences. It can detect known attacks, but it is not efficient when it comes to detect new or unknown threats like novel viruses.

1.2 Anomaly-based IDS

This is used to detect changes in behavior. The system is trained with a normalized baseline, which is basically a normal behavior. The activity is then compared against this baseline. If any event occurs that does not belong to this baseline, an alarm is triggered. This technique is efficient to detect unknown attacks.

These IDSs have proven that they are capable of preventing and protecting network segments. They are hence deployed on the networks in different machines and devices for protection against cyber threats. However, since nowadays much has become digital and network based, the rate of intrusions and their complexity has also increased. In such cases a single IDS proves to be very ineffective in protecting the whole network from attacks. A single IDS can easily let a sequence of attacks pass through (Wu et al., 2003). If we are not able to find the attack or virus at the proper time, it could cost major damage to the system and network. To overcome such issues, we should use collaborative intrusion detection systems/networks (CIDSs/CIDNs). Here, we see where we can use blockchain. The properties that blockchain exhibits are useful in the context of a CIDS. Specifically, the critical component of blockchain technology for CIDS applications is its mechanism of validating and storing data with no need for a central, trusted authority. As it possesses the unique feature of keeping data and information in different blocks and different systems using algorithms, it becomes difficult for an attacker to gain access to information because data is stored in different locations.

1.3 Organization of the paper

The remainder of this chapter is structured as follows. [Section 2](#) of this chapter explains IDSs and their functionalities. Along with this, it explains different forms of IDSs. In [Section 3](#), the blockchain is explained in detail. Also it discusses the reasons why blockchain is secure and reliable; this is due to the different consensus mechanisms that it uses to verify each transaction. [Section 4](#) focus on HIDS and its working. In [Section 5](#), blockchain-based

intrusion detection is explained as well as challenges related to the intrusion detection system and how blockchain helps IDS and is a solution to the challenges faced in IDS. [Section 6](#) focuses on a collaborative IDS and how blockchain can be a solution to problems like sharing of data and trust management in CIDS. [Section 7](#) discusses the applications of IDS, mainly Snort, and the designing of Snort-based CIDS with blockchain. [Section 8](#) deals with the limitations of both IDS and blockchain. [Section 9](#) illustrates how IDSs and firewalls, both being related to the security of the network, are different from each other. In [Section 10](#) is discussed the capabilities of blockchain and how much help it will be in the near future.

2. Intrusion detection system

The main objective of intrusion detection (ID) is to recognize any kind of virus or malicious activity that can be found in the host or in a network. There are various processes to approach the IDS, but [Fig. 5.1](#) shows the general process of an IDS. The figure shows the different parts of the process in how the system is capable of detecting an anomaly in the host or network ([Scarfone and Mell, 2007](#)).

ID gives details about the information gathered by monitoring the network and all kinds of system events with any incidents. Basically, an IDS can provide following functions.

2.1 Information gathering

As IDS monitors the system and network. It can collect all kinds of local information for the target. This information gathered can be processed and stored, which can further be used in managing the system ([Kanth, 2019](#)).

2.2 Generating the alert

The major role of an IDS is that it gives an alert to the information security administrator if any kind of attack or incident happens. It is necessary to keep a record at the time when it is attacked so it is effective ([Kanth, 2019](#)).

As we have seen, an IDS can be classified into NIDS or HIDS, so a network can be deployed on location like a wireless-based IDS that can



Figure 5.1 General process of an IDS.

monitor suspicious activity by monitoring wireless network packets through different kinds of protocols. Nowadays, IDS products use these two types of detection, providing more security and protection. The setup of such a network is shown in Fig. 5.2. There are five different forms of IDSs.

2.3 Network intrusion detection system

NIDSs are set up at specific points inside the network to analyze the traffic of data from all devices connected in the network. They carry the task of observing traffic on the subnet of the organization and matching the traffic data that is moved inside the network with a set of known cyber-attacks. Once an attack is recognized, an alert message can be sent to the security administrator. For example, NIDSs can be deployed on the firewall so we can notice if someone attacks it and be notified (Hu, 2010).

Once they are installed in the network, they start gathering information from a host console and check with the network for identification of permitted host computers, applications, and operating systems that are commonly used throughout the network. They also maintain a log on characteristics of normal traffic present in the network to identify any malicious activity or any changes in the network. They can also prevent different attacks like denial-of-service (DoS) attacks in a variety of ways, for example, by sending a TCP packet connection to prevent an attack, limiting bandwidth range and usage, or even by rejecting malicious activity or files present in the network. The setup of an NIDS is shown in Fig. 5.3 (Hu, 2010).

This gives various information of network traffic packets and different types of network protocols. The packets that are captured are analyzed by knowledge-based or behavioral-based approaches. In a knowledge-based approach, if packets match with a past record of attack, then it is an attack (Hu, 2010).

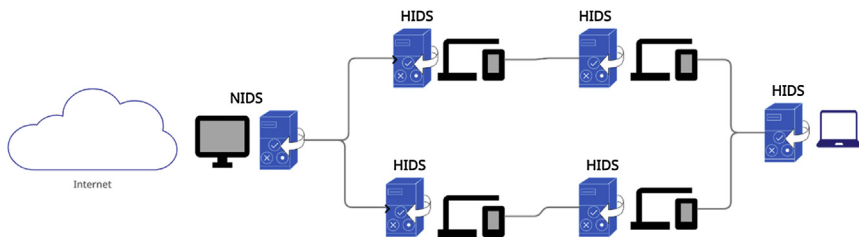


Figure 5.2 Network environment of HIDS and NIDS.

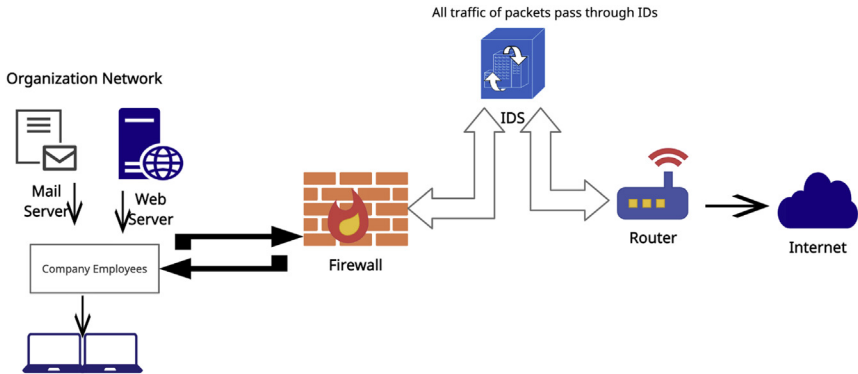


Figure 5.3 Network intrusion detection system.

2.4 Host intrusion detection system

An HIDS runs on each host or device inside the network. An HIDS keeps a record of all the incoming as well as outgoing packets from every device and notifies the security administrator in case there is any unwanted activity such as a malicious file or code being detected. A record of snapshots is maintained that consists of existing system snapshots and present system snapshots, so when any unwanted activity happens, it can compare these snapshots. If anyone tries to edit or delete the analytical system files, it will send an alert to the security administrator, so an investigation can be performed. For example, a laptop provided by a company to its employee for work may not have changes to any existing files or layout. The setup of an HIDS can be done as shown in Fig. 5.4 (Hu, 2010).

Regarding misuse, an HIDS can detect any kind of attack by the patterns of previous computer activities compared to current computer activities like CPU usage, memory used, and which files are access against the rules of host-based intrusions (Hu, 2010).

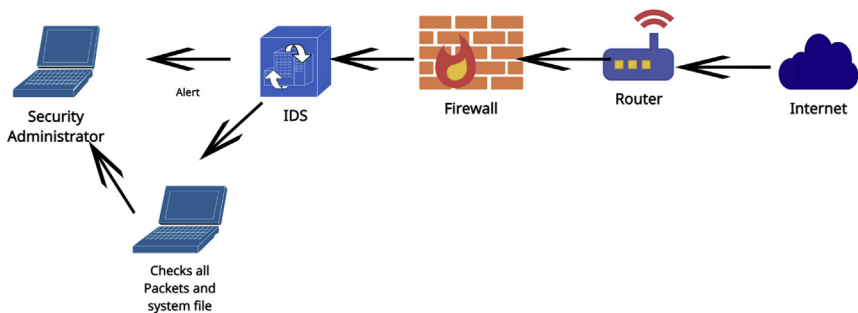


Figure 5.4 Host-based intrusion detection system.

2.5 Protocol-based intrusion detection system

PIDS is a device or sensor that detects the presence of an attacker trying to breach the physical parameter. It consists of a system that would consistently stay at the front end of a server; it is controlled and interprets protocols between a user device and the server, which can be a database. It mainly focuses on trying to secure the web server by keeping a record of monitoring the HTTPS protocol.

2.6 Application protocol-based intrusion detection system

It monitors and analyzes the important or specific applications present in a computer or device itself. Generally, this system is present within a group of servers. It monitors and interprets the application-specific protocols, thus identifying the intervention. For example, as it communicates with a database on the web server, this will track SQL protocol specifically displayed in middleware.

2.7 Hybrid intrusion detection system

As the name suggests, a hybrid intrusion detection system is made of two or more IDS techniques, e.g., Network intrusion detection system (NIDS) and HIDS. This is done to increase efficiency, as it allows detecting both known and unknown attacks. In this, a complete view of the network is created by combining host agent or system data with network information. This system is more effective in comparison to other IDSs.

3. About blockchain

Think of blockchain as a database with exception of its way of storing information and data. It is also referred to as distributed ledger technology (DLT) and follows peer-to-peer (p2p) topology. A DLT is a technology or system that stores digital data and records the transaction and its details in multiple places (servers) at the same time. So unlike traditional database systems, there is no central place where data is stored. Blockchain technology makes a transaction transparent by using decentralization, cryptography, and consensus mechanisms. Here, the digital assets are not copied or transferred. Instead, they are distributed over the network, and assets are decentralized, which means they can be accessed in real time.

The most common analogy to understand blockchain would be Google Docs. When we grant access to other people, the document is not copied; instead, it is distributed. If anyone makes changes to the document, it is

affected in real time. Other people do not have to wait while someone else is reading or writing. The same is the case with blockchain, but at a larger scale and more complex than this. In a world with blockchain technologies being followed, we can imagine that every transaction will be in digital code and will be stored in a shared database, decentralizing the network and making it accessible in real time, yet protecting it from alterations by hackers, deletion, or revisions. This is a world where bankers will no longer be required to identify, validate, store, and share payment information as it could all be done digitally using blockchain. The security aspects are discussed later in this chapter.

There are different types of blockchains based on your requirement. For example, if Bitcoin is considered, we need a blockchain that can be accessible by the public, so they can become one node and verify others and trade. But in the case of banks, the transactions need to be secured and only authorized members should have the right to access this confidential information. So in this scenario a blockchain that has limited authorization will be used. Different types of blockchain are detailed next.

3.1 Public blockchain

The public blockchain, also called a permissionless distributed ledger system, is a nonrestrictive blockchain system. Public blockchain systems are safe as long as participants follow security rules. A person can become an authorized participant by signing in on a blockchain platform. A participant or node can then access present as well as past records, verify a transaction by consensus mechanism, and do mining. These three are the most important tasks in blockchain, with mining and exchanging cryptocurrencies being the most basic use of a public blockchain. Examples include Bitcoin and Ethereum.

3.2 Private blockchain

A private blockchain, also called a permissioned blockchain, operates in a closed network. This type of blockchain is used in organizations where only a few people are allowed to be authorized to participate. This type of blockchain is the same as public but with more restriction and fewer users participating.

3.3 Consortium blockchain

This is a type of semi-decentralized blockchain that is managed by more than one blockchain network. It is like a permissioned blockchain. In this

type of blockchain, more than one organization can act as a node and exchange information or do mining. This type is mostly used in banks and government organizations.

3.4 Hybrid blockchain

This is a combination of public and private blockchains. This allows the user to control who can have permission to access data stored in the blockchain. Here a transaction in private mode can be verified in public.

Blockchain is not limited to cryptocurrency. It can be used to run websites using decentralized domains living on blockchain. Fig. 5.5 shows the blockchain of permissionless and permissioned, where an example is censoring the internet. A lot of countries hide information from their citizens about the outside world. VPN seems to be a solution, but governments often have the power to shut down VPNs. So a better solution is decentralized domains using blockchain, where websites are not hosted from only one location, so they are difficult to take down. This helps in fighting censorship.

3.5 Blockchain in security

Blockchain is a DLT, in which data are structured in the form of blocks and each block is chained together (Fig. 5.6). Blockchain is based on principles of cryptography, decentralization, and consensus, which ensures trust in transactions, a major limitation of IDSs.

Each block is connected to all the blocks before it in a cryptographic chain, as shown in Fig. 5.7. All the transactions within blocks are validated and agreed upon by consensus mechanism to ensure that each transaction is true

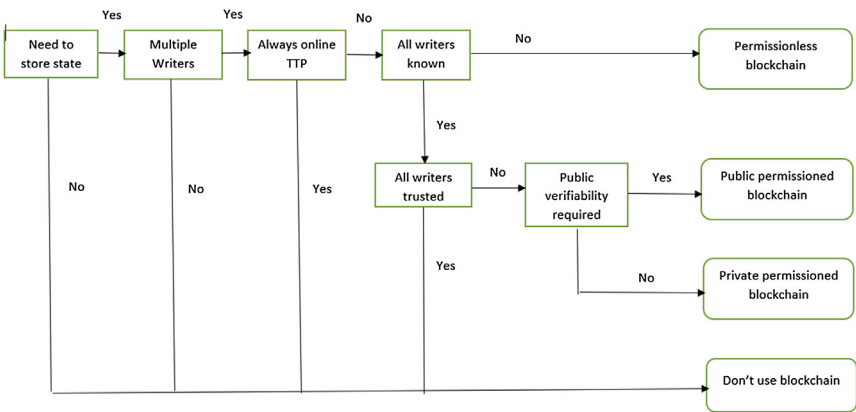


Figure 5.5 Permissionless and permissioned blockchain.

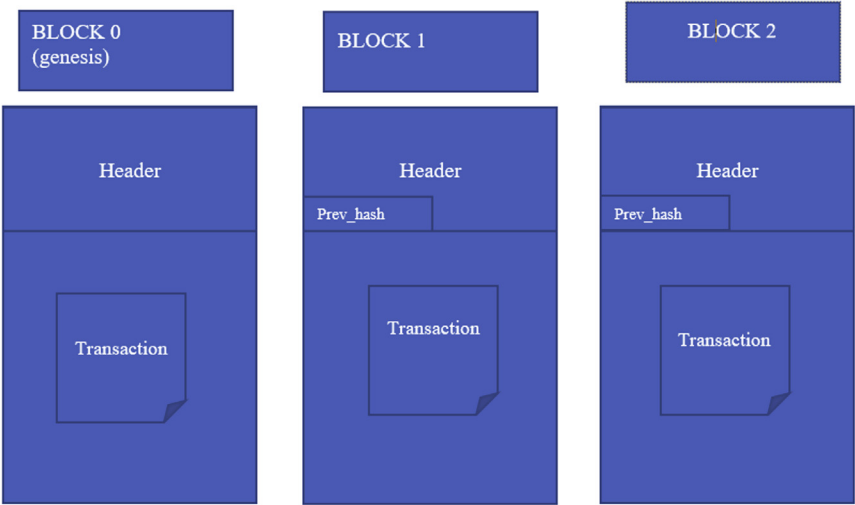


Figure 5.6 Blocks in blockchain.

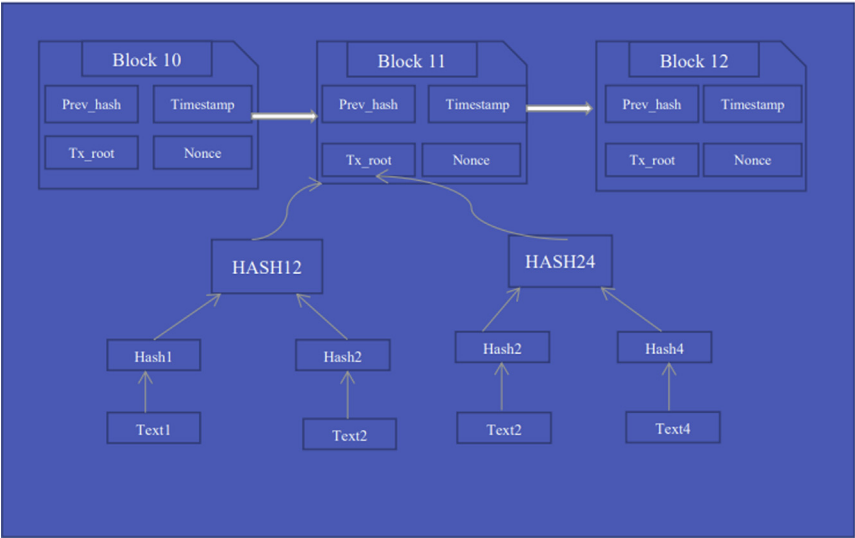


Figure 5.7 Blockchain structure.

and correct. Blockchain enables decentralization, which means members participate in a distributed network. This implies that a single user cannot change the record of a transaction, thus also providing security. The reason for this will be discussed subsequently. To understand the security concepts in blockchain, we need to understand its major components, presented next.

3.5.1 Block

Blockchain is composed of different blocks chained together. Each block has some basic elements:

- I. Data with nonce. Unless the data is mined, it is signed to a nonce and a hash. Nonce is a 32-bit random number, which is created when a block is generated. This nonce creates the block header hash.
- II. Hash is 256-bit number starting with many zeroes.

On creating the first block, the nonce creates a cryptographic hash.

3.5.2 Miners

Miners create new blocks on the chain through a process called mining. Each block consists of a nonce and hash, which is always unique, and it also references the previous block by hash. Basically, if you want to mine a new block, you have to change all the blocks, so mining is difficult. This provides safety to transactions. This is sometimes referred to as “safety in math.” To carry out mining, miners use special software.

3.5.3 Nodes

Decentralization is one of the most important concepts of blockchain. As we know, blockchain is a DLT, in which no one computer can own the chain. Instead the chain is connected by different nodes, where each node has its own copy of the blockchain. Nodes can be devices that keep copies of the blockchain and keep the network working.

Blockchains are transparent, so each transaction can be viewed and checked. Each participant has a unique identification number that shows its transaction. The network must check and approve newly mined blocks for the chain based on different algorithms. Blockchain provides integrity of transactions. This is achieved by cryptographic validation called hashing function, which links all blocks to one another in the chain. When any modification is done to the transaction in a block, it will cause the hash function in the next block to be invalidated, as each block has a hash function of the previous block. If a hacker wants to modify a transaction, he will have to mine the block and re-mine all other blocks that come after it, which is a tedious task and requires huge amounts of time and computing power. Integrity is one of the greatest merits of blockchain.

Blockchain provides reliability through verification of each transaction by consensus mechanism. Blockchain is considered secure and verified. This is possible because of consensus protocol, an important part of blockchain. A consensus algorithm is a procedure that all peers and nodes in the network

follow to reach common agreement about the state of the distributed ledger. Consensus protocol looks for every new block added to the blockchain by making sure that it is one along with the version that is agreed by all the nodes in the blockchain, thus, providing reliability and trust in the network. The consensus protocol has different objectives like coming to the same agreement, collaboration, cooperation, and providing equal rights to every node.

3.6 Consensus algorithms

3.6.1 Proof of work (PoW)

Bitcoin and Ethereum use this algorithm. The main use of this algorithm is to solve mathematical puzzles that are complex in nature. It is used to generate blocks by selecting the miner. This mathematical puzzle requires high computation power and enormous time. According to this algorithm the node that solves the puzzle first will mine the block.

3.6.2 Proof of stake (PoS)

This is a different approach to PoW. In PoW, to carry out verification, miners have to solve the mathematical puzzle. The first node to decrypt the puzzle gets the coin or to generate the next block. This requires high power. It has been approximated that one Bitcoin transaction will use enough energy to fuel 1.57 American households. Thus, to solve this issue, PoS attributed mining power (i.e., the ability of a miner or node to solve the mathematical puzzle) to a proportion of coins held by the miner. A block is validated in this way by putting a wager on whether or not it will be attached to the chain. Validators are awarded in proportion to their stakes based on actual blocks added, and their stakes are increased accordingly. Finally, a validator is selected based on its economic condition to add a new block to the network's chain. To reach an agreement in the case of PoS, an incentive scheme is used.

3.6.3 Practical Byzantine Fault Tolerance (PBFT)

This consensus algorithm focuses on providing a safeguard against system failures by collecting responses from correct as well as faulty nodes. This protocol is designed for asynchronous systems, to lower the overhead time. Its aim is to solve problems in a distributed network to reach a consensus even when nodes does not respond or respond incorrectly.

3.6.4 Proof of burn (PoB)

This algorithm is like setting priorities. Validators are required to sacrifice their long-term goals to achieve a short-term goal of mining the next block.

Here, validators “burn” their coins, which basically means that they have to send their coins to an IP address from where the coins cannot come back. The more they sacrifice (burn their coins), the more chance they have to be selected to mine the next block. This is a different approach from PoW.

3.6.5 Proof of capacity (PoC)

Rather than investing in costly hardware or burning, validators are expected to invest their hard drive space in this protocol. The more space they have, the more likely they are to be chosen to mine the next block.

3.6.6 Proof of elapsed time (PoET)

PoET is the most equitable algorithm since each validator has the opportunity to build their own block. Any node must wait for an unknown period of time before adding their proof of waiting time to the block, which is then connected to the network. The validator with the shortest waiting period is chosen, and his or her block is added to the blockchain.

3.7 Blockchain in IDS

In the last few years, many industries have been focusing on deploying blockchain-based applications. Most of these are used for some specific cases that have certain unique and particularly detailed blockchain requirements along with some uniquely detailed and customized characteristics.

For instance, there have been a lot of cases that were built on the basis of improving security and privacy among different divisions in a particular system. So, the major focus of such cases is to store the data completely privately (Zyskind et al., 2015). Similarly, most implementations use proprietary schemes with the cryptography application. This method ensures full privacy security and anonymity between different transactions (Li et al., 2013).

Technologies centered on blockchain are mostly used to store data in a decentralized way that can ensure security of the data. It is therefore of great importance to find out how blockchain systems can be deployed in IDSs.

Blockchain-based CIDN frameworks are designed for certain methods. In some the set of alarms that are not legitimate, also called false alarms, that are obtained from IDSs are considered blockchain transactions. In some cases, for safety against intrusions in blockchain, a unique communication protocol is used. This provides a stable transaction connection for the alarms being generated in a specific block (Alexopoulos et al., 2017). CIOta framework was also developed by some researchers to utilize the

methodologies of blockchain for collaborative anomaly detection (Golomb et al., 2018). Blockchain applications can also be protected with the help of IDS agents such as decision agent, monitor agent, manager agent, and action agent. These agents can be added to the security node for monitoring and analyzing purposes.

4. Host-based intrusion detection system

An HIDS helps us in detection of unauthorized use of the network and abnormal and malicious activities on the host. As we know, an IDS helps us in maintaining the security of a system by capturing and analyzing network traffic to detect suspicious activities (Axelsson, 2000; Debar et al., 1999; Holtz et al., 2011). An IDS works mainly on two approaches: anomaly detection and misuse/signature detection. We have different kinds of IDSs, as seen previously in the chapter. Based on the requirements of the organization, the kind of IDS that is to be used is selected. Sometimes, types of IDSs that work for different approaches are combined to get overall protection.

4.1 Working of an IDS in an HIDS

In organizations or anywhere where there is a large network, multiple NIDS are positioned and installed across the network. These systems share among themselves the log and alert information that they collect overtime. This type of arrangement and positioning is generally known as a distributed intrusion detection system (DIDS). The type of information or data and the amount of information to be shared among the systems is configured beforehand by the network administrator and should be modified and updated frequently according to the patterns of attacks being detected. These systems help the administrator get an overall view of network attacks (Ghribi et al., 2018). These IDSs help the administrators to analyze the threats and alerts of the whole network and then to recognize the complicated patterns of attack. Based on these attack patterns, certain signatures and sets of rules are created. They are then given to the IDS to secure the network segment from such attacks in the future (Titorenko and Frolov, 2018; Anwer et al., 2018; Kato and Klyuev, 2017).

These days the attack patterns have become more complex, and with the increasing variation in the size of networks, the number of attacks is also growing. To identify and prevent intruders and malicious activities, the administrator should know how the attack began, where it generated, what

the attacker did, and what the level of attack was. Because the whole framework is dependent on DIDS, it should be flexible and strong enough to detect the attacks quickly.

5. Blockchain-based intrusion detection

Attackers these days apply more complex, complicated, and advanced techniques to attack systems and to avoid getting detected. Misconfigured network segments can also compromise security given by the multiple IDSs that are connected to the centralized server and may result in giving un-authentic logs or alerts (Alexopoulos et al., 2017). So, to make the whole system trustworthy and robust, a reliable solution according to many researchers and industrialists is blockchain.

The following are some of the challenges that a DIDS faces, and we will see how blockchain can be used to solve them (Yang et al., 2018; Shvachko et al., 2010; Lee and Lee, 2012).

5.1 Challenges in distributed intrusion detection systems

5.1.1 Integrity

The first challenge is the integrity or the authenticity of the alert that is being generated by an IDS. For the smooth working of IDSs and proper protection of the network, it is of utmost importance that the logs and the alerts generated by the system are unmodifiable and inaccessible by any attacker.

5.1.2 Consensus

All the IDSs that are in the network should have a common set of rules about the type of alert and quality of the alert that is to be generated.

5.1.3 Scalability

For centralized servers, their computational demands should be altered according to the size of the network.

5.1.4 Privacy

Only the participating IDS should have control over the accessibility of data regarding which alert is generated. For the proper functioning of the whole system, it is also important for the central server to remain secure and reliable; it should not be affected by any intruders. The central server has the most important task of collecting all the logs and alerts from the IDS for easy analysis of data and decision-making. Hence, it is the most important challenge that should be satisfied (Meng et al., 2018; Kim et al., 2018).

5.2 Blockchain-based solution

The way blockchain could be implemented is by having a distributed ledger that is secured and protected for the purpose of sharing of logs and alerts that are generated by individual IDSs. The alerts that are generated by the IDS will then be stored in the blockchain and are called transactions. The IDSs that are in connection with the centralized server then run a protocol that validates all the transactions that occurred. Only after validation is the transaction added to the blockchain, as shown Fig. 5.8. This kind of arrangement acts as a guarantee that only the alerts that are authenticated and validated will be updated in the server. These alerts cannot be modified and are visible to all the IDSs present in the network (Foroushani and Yue, 2018; Titorenko and Frolov, 2018).

The diagram shows the transfer of alerts to the cloud-based centralized server, using the alert exchange layer. The alerts are then validated by the blockchain consensus layer, and if their authenticity is proved the transaction is updated in the blockchain of all the IDSs. In Fig. 5.8, all the IDSs are connected to a DIDS that acts as a regulator between all the IDSs present in the network. Whenever any modification takes place in any of the IDSs in the network, an alert is generated, which goes to the distributed IDSs. This checks the authenticity of the generated alert and makes sure that the alert generated is genuine and is not the result of any malicious

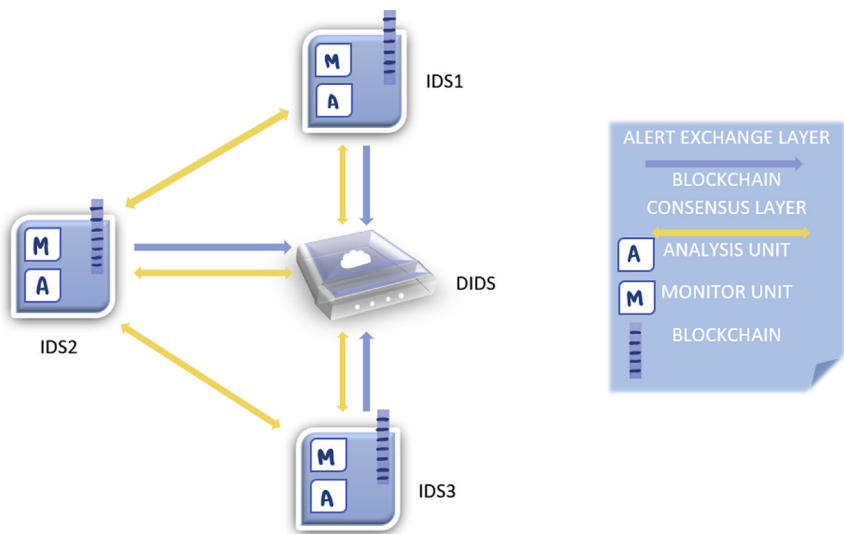


Figure 5.8 DIDS architecture using blockchain and cloud.

activity. Only after the authentication and validation of the generated alert will the DIDS allow the changes to be saved and updated in the blockchain. The figure thus shows how the DIDS regulates the activities that occur in any of the IDSs in the network and makes sure that all the IDSs in the network have up-to-date data in the respective blockchain.

5.3 Why a cloud-based centralized server?

Different scenarios may require different sized networks. It is important to maintain the performance and scalability of a DIDS even if the network is really large. In large networks, the load on the centralized sever will increase as all the IDSs dump their logs to it, and it becomes big data. To solve the challenge of analyzing such large data in real time, we need a cloud-based centralized server that is cost-effective and scalable.

6. Collaborative intrusion detection system

The discovery of collaborative intrusion detection involving CIDNs and CIDSs was carried out practically. Performance improvement was the main objective that could be surpassed easily by complex attacks like DoS attacks. The main reason is that IDSs usually do not have information about their protected area. The framework of a CIDS allows the various areas of the IDS to understand the context by data exchanging with one another. Collaborative programs can be classified into these types:

- (a) hierarchic systems such as DIDS (Snapp et al., 1991) and EMERALD (Porras and Neumann, 1997)
- (b) subscribe systems such as Distributed Overlay for Monitoring InterNet Outbreaks (DOMINO) (Barford et al., 2004) and COSSACK (Papadopoulos et al., 2003)
- (c) P2P querying-based systems like Netbait (Chun et al., 2003) and PIER (Huebsch et al., 2005)

Event Monitoring Enabling Responses to Problems with Anomalous Live Disturbances (EMERALD) (Porras and Neumann, 1997) is detection of malicious incidents in a layer of layers that are not visible on a large network. Similarly, DIDS was developed (Snapp et al., 1991). It pointed out disadvantages by combining centralized data analysis, data reduction, and distributed monitoring. Papadopoulos et al. founded COSSACK (Papadopoulos et al., 2003), which focused on zero human intervention to intelligently reduce distributed denial-of-service (DDoS) attacks. Yegneswaran et al. founded DOMINO (Barford et al., 2004), which can

increase the effectiveness of detection by directing interactions between illegal sites or heterogeneous nodes. PIER (Huebsch et al., 2005) is an internet-scale query engine that can support the most widely distributed and ongoing queries and can be used as a platform to build a set of information centric applications.

In Fig. 5.9 the architecture of a CIDN is shown. In this, a node, such as node A, can exchange data needed for one another node (such as nodes B, C, D). A node is often made up of many parts or components: a collaboration component, P2P communication, and an IDS module component. Especially, functions of intrusion detection can be performed by the IDS module, including functions such as recording events and network traffic monitoring. The component of collaboration is responsible for helping the node in exchanging required data with other nodes and performing specific tasks such as trust computation. The objective of the P2P communication component is to help in establishing a physical communication with another IDS node.

6.1 Working model of CIDS

The elementary detectors (EDs), the message queue (MQ), the connection tracker, the manager, and the response engine are the components of the CIDS operating model. The EDs are specialized intrusion detectors dispersed throughout the system. To communicate with one another, all of the system's components use MQ, which employs TCP as the data

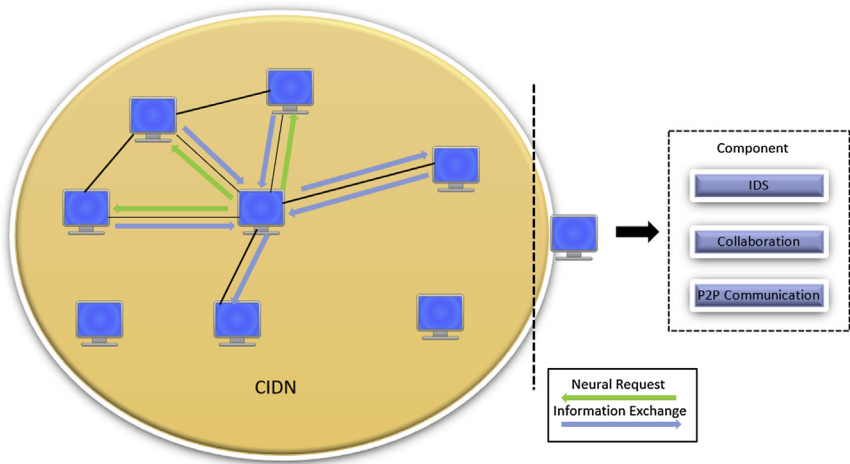


Figure 5.9 CIDN architecture.

transport. The connection tracker is a kernel-level entity that keeps track of the mapping of a port number to the process ID of the process that is currently connected to that port. The manager is the workhorse, and it is in charge of combining data from several detectors and making a system-wide decision in the presence of an intrusion.

6.2 Collaborative intrusion detection challenges

Studies on ID have been done for approximately 40 years, and the conclusions show that sharing of data and the computation of trust in collaborative environments remain two major challenges.

6.2.1 Sharing of data

Sharing of data is one of the major problems of the collaborative process because it is not a small or easy task to allow all participants to trust one another. For instance, some amount of trust can be built by the help of Public Key Infrastructure (PKI) technology, but it does not work every time in terms of detecting intrusion. In addition, due to some concerns of privacy, some participants are reluctant to pass or share information. Without sufficient data, they cannot perform algorithms of detection and create a model for the detection of suspicious or malicious events.

6.2.2 Management of trust

It is evident that CIDNs/CIDSs are at risk of internal attacks, where intruders authorize access to the network. Typically, computer dependencies are usually used to measure levels of trust between other nodes. In doing so, a server (central) is set up to collect traffic data and node behavior and calculate the value of trust for each and every node. Management of trust will be a problem when an organization is large because it is very difficult to find a third party that is reliable; that is, the server (central) can be compromised.

6.3 Solution for CIDs using blockchain

6.3.1 Sharing of data

The problem of sharing of data is mainly due to two things: trust and confidentiality of data. The term mutual trust means that when data or information is shared, working groups should have trust that others will not disclose any information. For example, two IT organizations would make an agreement that they will not share or disclose any data with other parties. Privacy of data indicates that the shared data may contain specific

information related to a real company, i.e., traffic shared including packet payloads and IP addresses that could be used to transfer organizational privacy.

Blockchain can be used as a solution to this problem. Especially, sharing of data can be taken as a series of transactions. First, the groups must form an agreement for sharing of data, and each group should digitally sign it. After that, a blockchain box agreement can be kept that is consistent and public. In this case, other groups can also access the box of blockchain, read the agreement, and verify the data identity. The constant appearance of the agreement ensures that one group cannot reject it by disagreement. It is similar to using blockchain in the healthcare sector (Sotos and Houlding, 2017). An open accounting system is able to provide trust between the many working groups.

6.3.2 Trust computation

Generally, centralized, hierarchic, and distributed are the types of collaborative network architecture. In the literature, structures that are distributed have been widely discussed, whereas the two others are believed to have a problem of failure and a problem of one point of failure. In CIDN, warning exchanges are very important between the different IDS nodes, which can determine if anything is wrong.

The technology of blockchain can help to reduce this problem. For example, a blockchain-based CIDS that was introduced by Alexopoulos et al. (2017) uses blockchain so trust can be increased between IDS nodes. They view the alerts (raw alerts) generated by each IDS dot as a transaction of blockchain, which can be repeated between CIDN affiliate sites. After that, all collaborative nodes accept a sequence of rules to verify the authenticity of the functions before placing them on board. This function can ensure that warnings stored in the blockchain are disputed.

7. Applications of IDS: Snort

Snort is an open-source software. It is used for IDS or Intrusion Prevention Systems (IPS). It can perform real-time traffic analysis on IP. It has other features like protocol analysis, searching the content and matching it over the network-based system. The Snort program has the ability to detect a variety of attacks, such as OS fingerprinting, semantic URL attacks, buffer overflows, server message block probes, and stealth port scans, among others.

We can configure Snort into three different modes:

7.1 Sniffer mode

In sniffer mode, it will run a program that will scan the packets present in the network that will be shown on the console.

7.2 Packet logger mode

In this mode, it will run the program to log packets present in the network to a storage device like disk, hard drive, or pen drive.

7.3 NIDS mode

In this mode, the running software has the ability to track packets in network traffic and analyze them using a set of rules specified by the user. The running software will then take a specific action in response to what has been found.

7.4 Designing a Snort-based CID with blockchain

This section highlights the deployment and extensive use of software-defined networking (SDN). The model has a blockchain-based CID-based network topology. Snort IDS is used as an HIDS and a Ryu versatile network-based NIDS. The main goal is to defend against attacks that are performed internally and to improve the detection of attacks and their accuracy with the help of blockchain through a Snort node between the control plane of the application and the entire node's signature database. As a unique hash function, they compiled and connected the entire job's ID. This hash number is used by each Snort node, and the new ID is used for delivery. Similarly, for the next Snort position in the current blockchain, the hash value of a block and transaction ID of the last used block will then be used as the hash value for new blocks (Karupiah and Saravanan, 2014).

Fig. 5.10 depicts information of the blockchain network between CIDN nodes. Here Snort1 to Snort2 and Snort2 to Snort3 nodes can interact with each other with the help of a hash table that has the public and private keys. In this network topology, Snort1 node is connected to Snort2 by hash value, and Snort2 is connected with Snort3 in the same manner. In this way, Snort nodes can communicate with each other by assigned transaction (Karupiah and Saravanan, 2014).

This system can detect various kinds of DDoS attacks, which are the following: DDoS floods, SSH attacks (this represents SSH exploits), FTP attacks such as brute force type, HTTP attacks (which perform HTTP floods on the attacker's system), ICMP attacks such as Smurf, ARP attacks (which perform

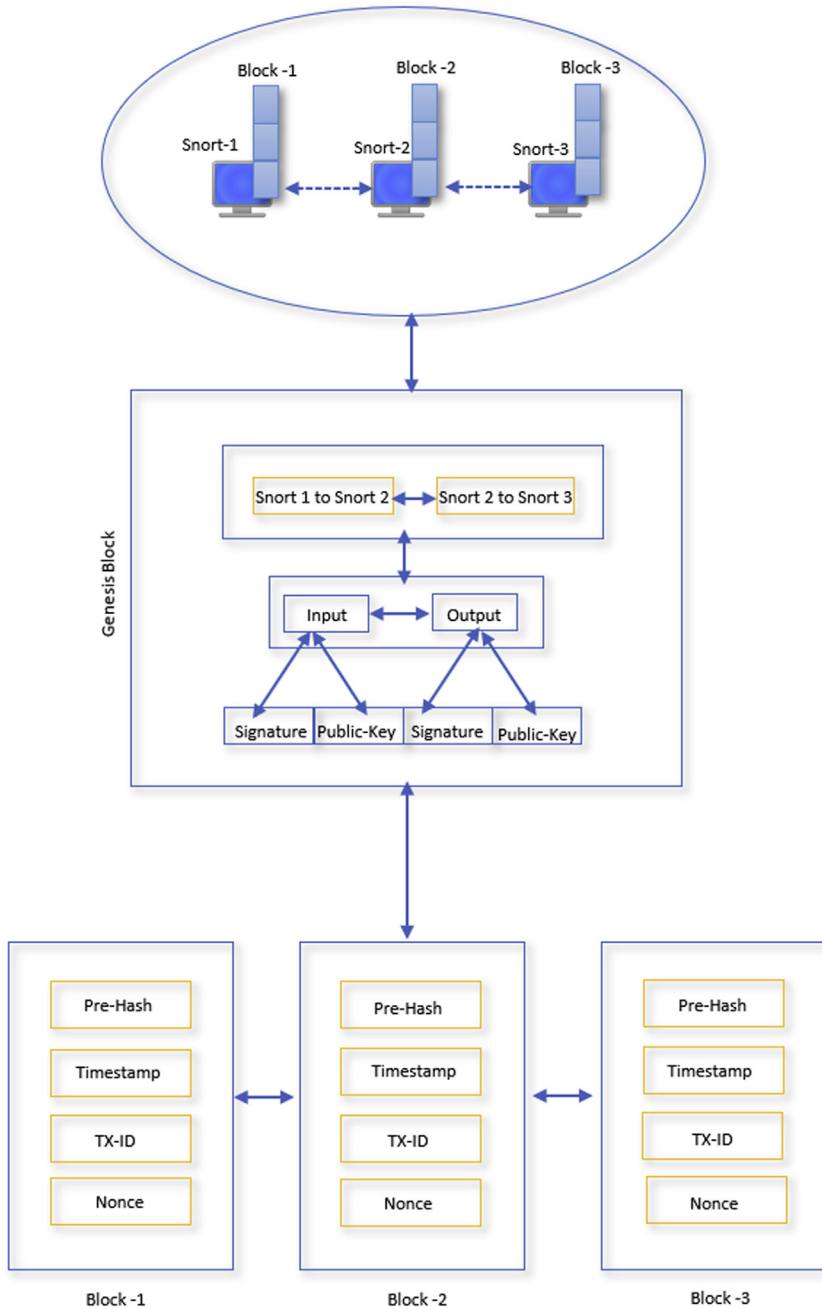


Figure 5.10 Blockchain trust communication inside CIDN Snort nodes.

ARP spoofing), and scan attacks that scan all ports. Their design consists of three different components: Ryu controller, IDS nodes, and blockchain certificates trust, as shown in Fig. 5.11. It has the flexibility of three VM's, like VM-1 that contains the integration of SDN Ryu and Snort and VM-2 that depicts the network domain, in which network emulation is used to simulate a virtual CIDN. There are three Snort IDS locations on the CIDN. Direct attacks on CIDN-1 and CIDN-2 nodes are launched using VM-3. Only a capture adapter is used for this virtual machine (Ujjan et al., 2019).

7.5 Evaluation of this design

The proposed system's performance in an SDN-based environment uses the CIDN to measure the collective-based Snort CDN performance of the

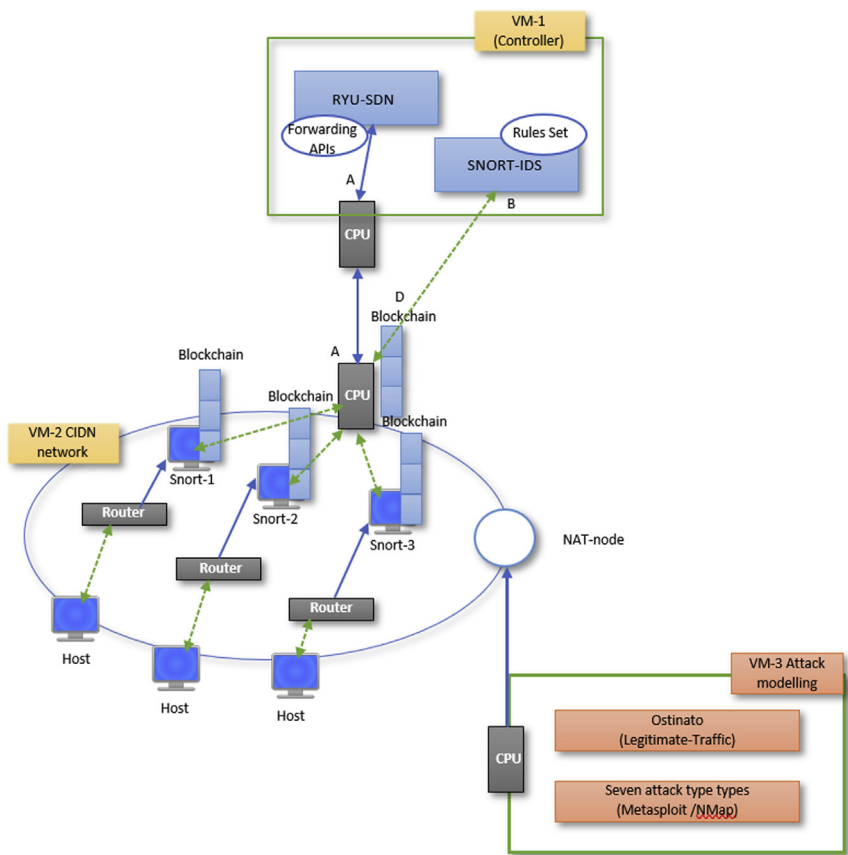


Figure 5.11 Snort IDS designed in SDN using three VM's.

network with implementation of blockchain. It makes use of blockchain in Snort nodes, which track, detect, and share the rule set with all neighboring nodes that are present in the Snort signature (Ujjan et al., 2019).

7.5.1 Attack modeling

HTTP, FTP, and SSH resources were used on VM-3 as a server for attack modeling. Seven different types of malicious traffic were created by using Metasploit and Kali Linux. The intention was to generate malicious traffic. CIDN IDS positions were inserted into all major attacks and the official route. All blockchain-based IDSs began testing dangerous and permitted traffic and issued warnings when input traffic matched a rule set by blockchain for Snort IDS, causing network traffic to be disrupted (Ujjan et al., 2019).

7.5.2 Evaluation with poisonous and permitted traffic in a CIDN

The output of a Snort IDS that has been collaborated with a blockchain-based CIDN has been investigated. The CIDN was used to look at the output of malicious and permitted traffic. There were three Snort IDS nodes with blockchain collaboration. Each node was also connected to an SDN-based switch. In the CIDN, there were seven different types of malicious attacks, with each node employing the Snort signature rule set. To test the CIDN detection accuracy, we only used three nodes in the network. Metrics for IDS results are given (Ujjan et al., 2019).

1. true positive (TP)
2. true negative (TN)
3. false positive (FP)
4. false negative (FN)

Snort IDS nodes that collaborated in CIDN had a default set of rules. They used seven different attack combinations with permitted traffic. Using a trusted blockchain, they allowed a collaborated IDS to get the set of rules from the Ryu controller in the period's second run. The proposed developed model's overall detection efficiency was assessed using a blockchain-based backend framework that allowed each IDS node to incorporate new signatures from Ryu, allowing for more accurate and efficient detection (Ujjan et al., 2019).

7.5.3 Analysis of a CIDN with permitted traffic

The setup used a CIDN that comprised three Snort IDS used for designing a Snort CID. In this, Snort nodes used blockchain in collaboration with the

nodes. To see how the proposed idea of design worked, they watched when all Snort IDS were put together and analyzed their performance with the help of permitted traffic, which they created. The utilization of the CPU in the CIDN with blockchain when all Snort IDS were together was analyzed and contained three Snort IDSs nodes, Snortnode1, Snortnode2, Snortnode3, each of which used a shared processor unit rather than the entire CPU. Furthermore, since Snort is a signature-based IDS, it has limited processing power when it used as a standalone IDS during periods of high network traffic. As a result, the CIDN network was able to correctly detect intrusions with a low number of false triggers. A Snort IDS can also accept new indications from the Ryu controller due to the blockchain-based approach. Overall, we can observe that when they integrated all Snort nodes, the proposed design of the system's processing of packets is feasibly enhanced, allowing it to capture different forms of attacks quickly (Ujjan et al., 2019).

8. Limitations

8.1 IDS limitations

Studies on ID have been for a very long time, but still there are various unresolved problems in applications of real time that could significantly reduce detection function (Wang and Liu, 2015).

8.1.1 *Traffic overhead with handling capability limited*

Overload packets will dramatically reduce the detection system's efficiency in high-traffic networks. A significant number of network packets should be discarded if traffic exceeds the IDS processing power. For example, the computational burden is proportional to the size of the packet payload.

8.1.2 *Signature coverage in limited range*

The capability of detection when signature-based relies largely on signatures that are accessible. In another way it can be said that the number and quality of signatures sent determine detection results. Signatures, on the other hand, are usually restricted and cannot cover all reported attacks and abuses.

8.1.3 *Incorrect profile establishment*

With detection that is anomaly-based, it is very difficult to create a standard correct profile as it has strong traffic. Specifically, IDSs that are anomaly-

based many times use ML methods for creating a profile. Data training, specifically that with data attacks that are severely limited within performance, leads to incorrect classification of ML.

8.1.4 Major false warnings

It is important for the IDS to give timely alerts to security managers when there is a problem with the network. False alarms, on the other hand, are a significant problem during detection due to immature signatures and incorrect profiles, which can significantly reduce detection efficiency and add to security analyst workload. For example, a big organization may produce over 10,000 fake warnings in a day.

8.2 Blockchain limitations

As summarized in [Fig. 5.12](#), blockchain technology is a modern approach that still faces some problems and limitations.

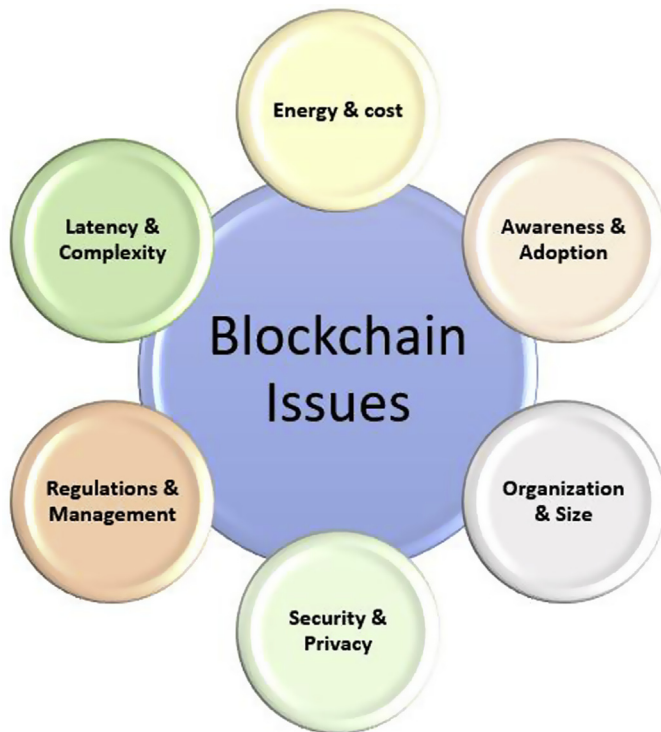


Figure 5.12 Blockchain technology: challenges and limitations.

8.2.1 Cost and energy

The power of computation is concerning with the usage of blockchain (Yli-Huumo et al., 2016). Take Bitcoin mines as an example: they require a high level of power to validate and calculate transactions. Liu and Wang (Wang and Liu, 2015) found that power of computation was added to the individual miner initially, but it could increase significantly as the network changed.

8.2.2 Security and privacy

Applications related to blockchain to be linked to known brands require smart transactions and contacts, which raises privacy concerns about data stored in a ledger that is shared. In addition, the technology of blockchain itself can be an attraction for cyber criminals, so it can suffer from many attacks such as DDoS (Karuppiiah and Saravanan, 2014, 2015; Karuppiiah et al., 2016, 2017, 2019; Kumari et al., 2016; Naeem et al., 2020; Maria et al., 2021; Pradhan et al., 2018; Li et al., 2017).

8.2.3 Complexity and latency

As we know, with the distributed nature of blockchain transactions, they will take several hours to complete before all parties have reviewed the corresponding ledgers. This delay could trigger a great deal of confusion among business partners, as well as provide an incentive for cyber criminals.

8.2.4 Awareness and adoption

The lack of understanding and acceptance of blockchain technology is one of the most important challenges. For example, many people need to know how it is useful and secure. The blockchain's potential growth is dependent on how many teams implement the technology, but this is still an open question.

8.2.5 Size and organization

Many different organizations use their own standards to create blockchains. The increased size of distributed ledgers could lead to significant reduction of performance and compared to current frameworks make blockchain less efficient.

8.2.6 Management and regulations

In advanced technology, regulations are usually far behind. For completing transactions on blockchain, due to the lack of standards and for better efficiency, Bitcoin blockchain has exceeded existing regulations. Applications of blockchain are expected to work within the rules.

8.3 General IDS limitations

- The effectiveness of IDSs can be severely affected by noise. The bad packets that are generated from software bugs, corrupted DNS data, and the local packets that somehow escaped create a very high rate of false alarms, thereby reducing the effectiveness of the system.
- In most cases the number of real attacks is lower, especially when compared to the number of times the alarm goes off. Hence, most alarms are false alarms, and a lot of energy and time can be spent in categorizing the attack as real or false, leading to a real attack being missed or ignored.
- The library of signatures and the set of rules should be changed and updated as regularly as possible to mitigate and reduce the threats. Many times, the library might get outdated with only certain types of attack patterns stored in it or they might be only for a specific version of software. These outdated databases can make the system more vulnerable to attacks that use newer methodologies and techniques
- In the case of signature-based IDSs, there often is a lag between discovering a new threat and then applying the signatures generated to the IDS, so during this time the IDS may become vulnerable to similar attacks.
- IDS is completely dependent on the signatures and the set of rules that are provided to it, so if any attacker gets into the network even after the IDS is configured properly, the IDS can do nothing to prevent the malpractice. Therefore, IDS cannot compensate for a poor set of network protocols or weak authentication methodologies used.
- An encrypted packet that is not discovered until more significant network intrusions are detected can intrude the network and will not be identified by the IDS.
- Using IDS in networks will only be beneficial if the IP address of the packet being analyzed and monitored is correct, as the IDS works only on the basis of the network address that is being provided to it by the packets. If the IP address of the packets is spoofed or is falsified, the IDS will not be able to protect the network from the intruder.
- As NIDS are needed to analyse the packets as soon as they are captured, they can be susceptible to the same attacks to which the network host is vulnerable. The NIDS may also crash due to attacks by invalid data and TCP/IP.

9. Comparison with firewalls

Though IDSs and firewalls are both related to security of the network, IDS is different from a firewall in terms of a traditional network firewall; that is, different from a next-generation network, to allow or deny any network communication it uses the same rules. It completely blocks entry by assuming the correct rules are set. In fact, to prevent intrusion and not signaling of attacks from within the network, the firewall tries to restrict access between networks. Once intrusion has occurred, the IDS describes the suspected intrusion and signs the alarm. Attacks from within the system are also monitored by the IDS. They are detected by examining communications of networks and identifying patterns (which are also known as signatures) of common attacks on computers, as well as taking steps to notify operators. An IPS is the system that terminates the connection and performs access control like a firewall of an application layer.

10. Conclusion and future scope

With such capabilities, it is expected that blockchain will definitely emerge across different industries and domains like IOT, machine learning etc. With more use of proof-of-concept, the technology will validate itself. When combined with ID the technology will make huge impacts in possibly every digital transaction in the world and impact trade-offs and costs. A major application would be sharing of data, which is a huge problem for large, distributed organizations as performance, trust, and data privacy need to be improved. There are not a lot of real-time exchange applications available, so data sharing and alert exchange need to be improved. Blockchain is an emerging technology in field of data management. It is unlike any traditional data management system, where real-time information exchange, decentralization, and distributed ledgers are not available. Blockchain for ID is one of the trending areas of research and has a huge potential to affect the world. Our chapter focused on understanding the concepts of blockchain regarding ways in which it helps in security and IDSs.

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